3.3 Derivatives of the Inverse Trigonometric Functions

Name _____ Period ____ Date _____

AP Calculus 3.3 · Student sheet

I can…

- find the derivative of an inverse function at a point using $g'(b) = 1/f'(a)$.
- find the derivatives of $\sin^{-1}x$, $\cos^{-1}x$, and $\tan^{-1}x$.
- use the Chain Rule with inverse trig functions.

Vocabulary · fill in the missing word

Inverse function — $g =$ _____ $= x$; if $f(a) = b$ then $g(b) = a$

Derivative of an inverse — $g'(b) =$ _____, where $f(a) = b$ and $f'(a) \neq 0$

Derivative of $\tan^{-1}x$ — $\frac{d}{dx}\tan^{-1}x =$ _____

Derivative of $\sin^{-1}x$ — $\frac{d}{dx}\sin^{-1}x =$ _____, $-1 < x < 1$

Part 1 · Mirror slopes

$f(x) = x^3$ and $g(x) = \sqrt[3]{x}$ are inverses.

Point	Slope
f at (2, 8)	
g at (8, 2)	
How are they related?	

Part 2 · Notes and examples (fill in each note, then work the example)

Derivative of an inverse: if $f(a) = b$, then $(f^{-1})'(b) =$ _____.

Example 1

$f(x) = x^3 + x + 1$ and $g = f^{-1}$. Find $g'(3)$.

Example 2

$f(2) = 7$ and $f'(2) = -5$. $g = f^{-1}$. Find $g'(7)$.

Inverse sine and cosine: $\frac{d}{dx}\sin^{-1}x = 1/\sqrt{1 - x^2}$ and $\frac{d}{dx}\cos^{-1}x =$ _____.

Example 3

$f(x) = \sin^{-1}x$. Find $f'\left(\frac{1}{2}\right)$. Then find $\frac{d}{dx}\cos^{-1}x$ at $x = 0$.

Inverse tangent: $\frac{d}{dx}\tan^{-1}x =$ _____.

Example 4

$f(x) = \tan^{-1}x$. Find $f'(\sqrt{3})$ and $f'(-1)$.

Chain Rule: $\frac{d}{dx}\tan^{-1}u = u'/(1+u^2)$ and $\frac{d}{dx}\sin^{-1}u =$ _____.

Example 5

Find each derivative.

a. $y = \tan^{-1}(3x)$

b. $y = \sin^{-1}(x^2)$

Part 3 · Partner practice

Partner 1

$f(x) = \tan^{-1}(2x)$. Find $f'(1)$.

Partner 2

(AP® Practice) Let $f(x) = x^3 + 2x - 1$ and $g = f^{-1}$. Then $g'(2) =$

(A) $\frac{1}{5}$ (B) 5 (C) $\frac{1}{14}$ (D) 14